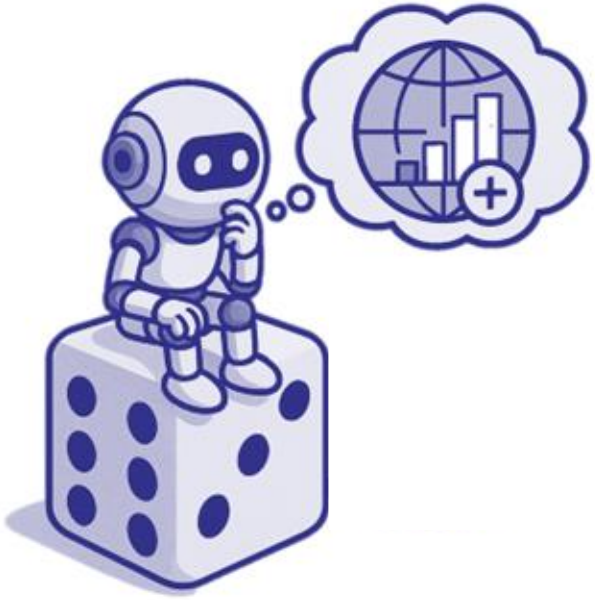
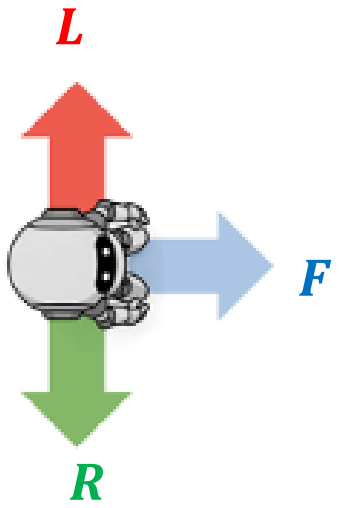
Discrete Probability

The Robot on a Die Example





- A robot lives on die world.
- Its position is one of the six faces.
- State: $x_k \in \{1, 2, 3, 4, 5, 6\}$
- Motion Command: $u_k \in \{F, L, R\}$
- **Never rotates**
- Observation: $z_k \in \{Even, Odd\}$

- Motion model: $\bar{x}_k = f(x_{k-1}, u_k)$
- Observation Model: $z_k = parity(x_k)$

The Robot on a Die Motion Model

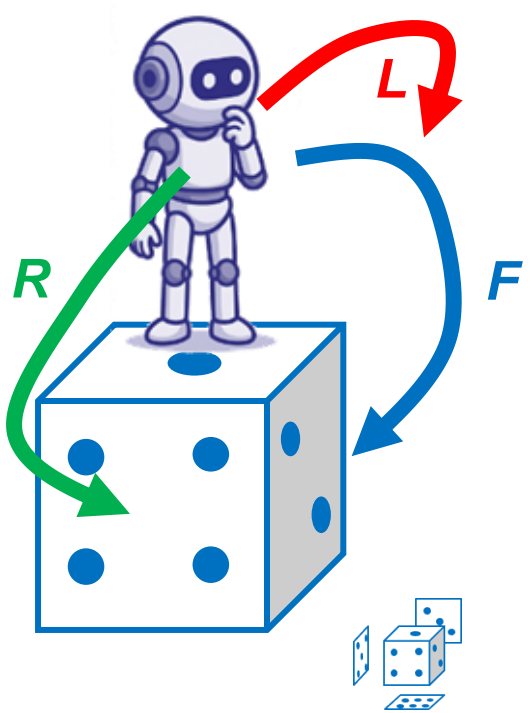
Successful Motions:

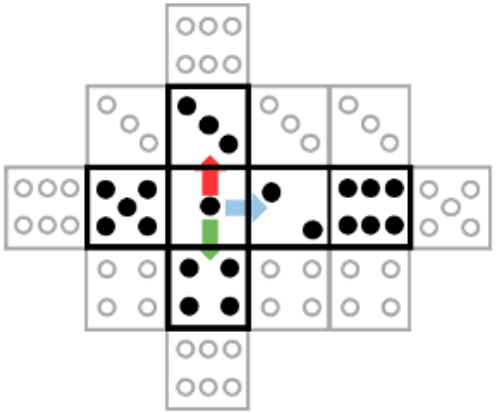
- From current face to adjacent Face in the motion direcction
- P=0.5

Failure Motions

- From current face to any non adjacent face in the motion direction
- P=0.1

$$f(x_{k-1} = i, u_k = F) \Rightarrow \begin{cases} P(\bar{x}_k = adj(F, i)) = 0.5 \\ P(\bar{x}_k \notin adj(F, i) = 0.1 \end{cases}$$

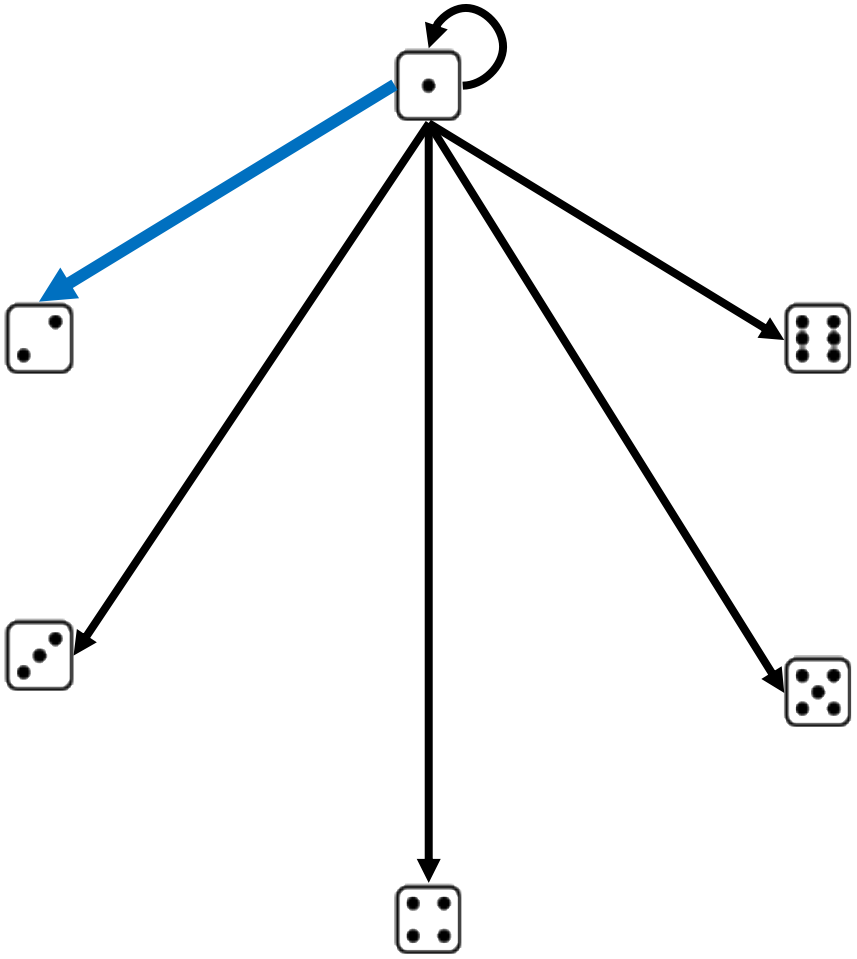




adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4

The Robot on a Die Motion Model



Transition diagram for:

- $x_{k-1} = 1$
- $u_k = F$

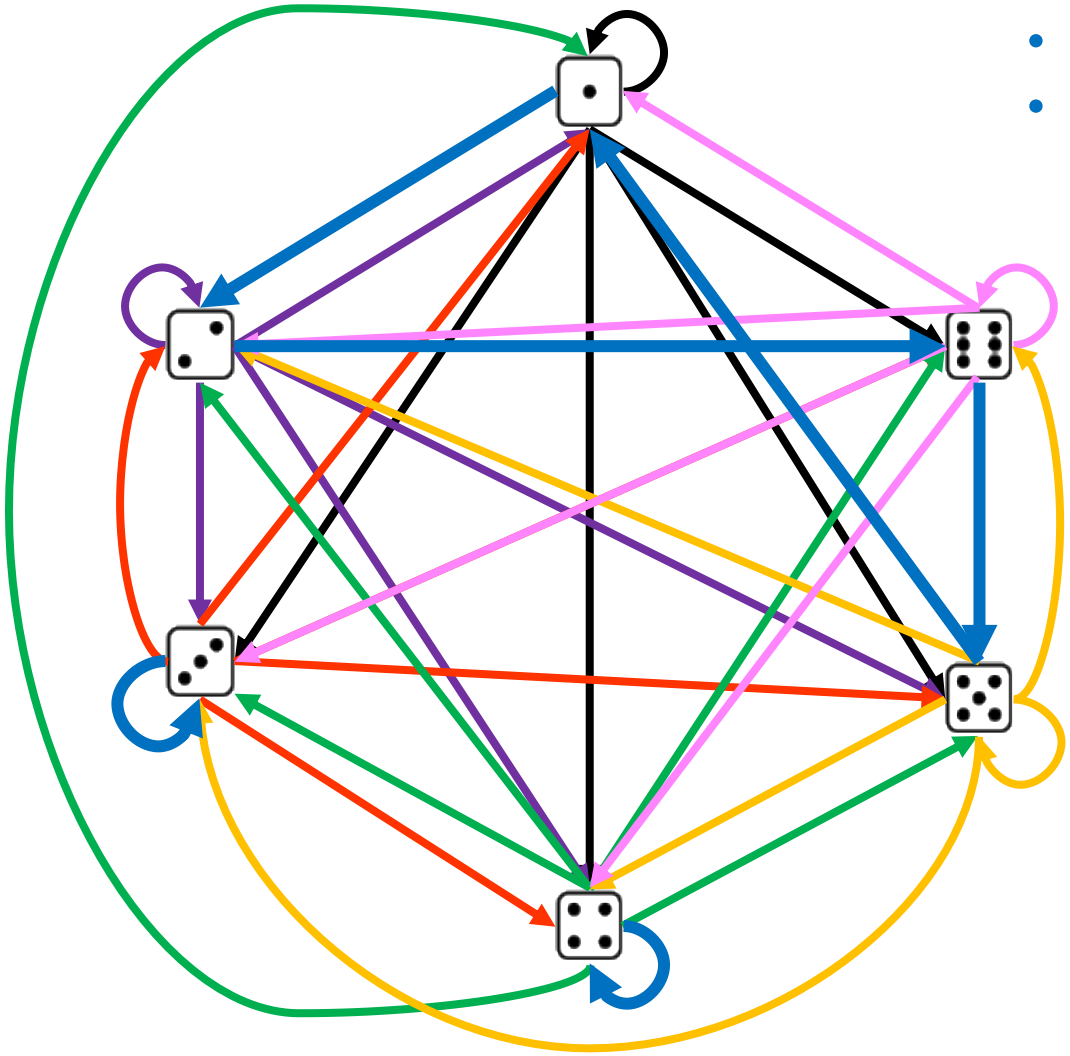
adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4

Successful F motion

Failure F motion from ⚀

The Robot on a Die Motion Model



Transition diagram for:

- $x_{k-1} \in \{1, 2, 3, 4, 5, 6\}$
- $u_k = F$

adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4

Successful F motion

Failure F motion from ⚀

Failure F motion from ⚁

Failure F motion from ⚂

Failure F motion from ⚃

Failure F motion from ⚄

Failure F motion from ⚅

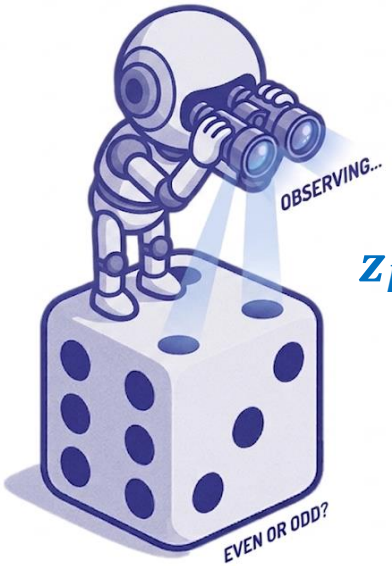
The Robot on a Die: Observation Model

Successful Observations:

- The sensor returns the correct parity of the face
- P=0.9

Failure Observations:

- The sensor returns the wrong parity of the face
- P=0.1

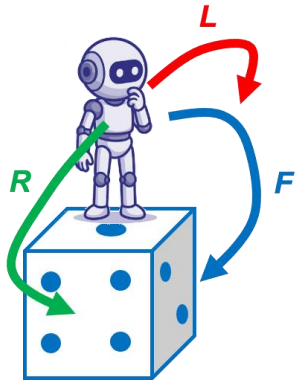


$$z_k = parity(x_k) = \begin{cases} x_k \in \{1,3,5\} \Rightarrow \begin{cases} P(z_k = Odd) = 0.9 \\ P(z_k = Even) = 0.1 \end{cases} \\ x_k \in \{2,4,6\} \Rightarrow \begin{cases} P(z_k = Odd) = 0.1 \\ P(z_k = Even) = 0.9 \end{cases} \end{cases}$$

Discrete Probability

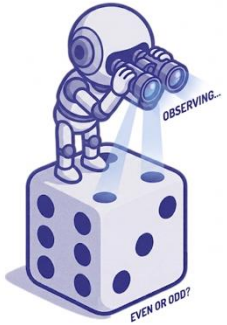
Random Experiment

Definition 3.2.1 — Random Experiment. A *random experiment* is a process or procedure that, when performed under specified conditions, produces one outcome from a set of possible outcomes. Although the set of possible outcomes is known in advance, the particular outcome that will occur cannot be predicted with certainty before the experiment is carried out.



Example 1: Moving is a random experiment

$$f(x_{k-1} = i, u_k = F) \Rightarrow \begin{cases} P(\overline{x}_k = adj(F, i)) = 0.5 \\ P(\overline{x}_k \notin adj(F, i) = 0.1 \end{cases}$$

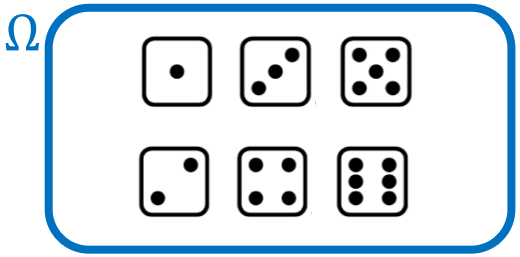


Example 2: Parity Observation is a random experiment

$$z_k = parity(x_k) = \begin{cases} x_k \in \{1, 3, 5\} \Rightarrow \begin{cases} P(z_k = Odd) = 0.9 \\ P(z_k = Even) = 0.1 \end{cases} \\ x_k \in \{2, 4, 6\} \Rightarrow \begin{cases} P(z_k = Odd) = 0.1 \\ P(z_k = Even) = 0.9 \end{cases} \end{cases}$$

Sample Space

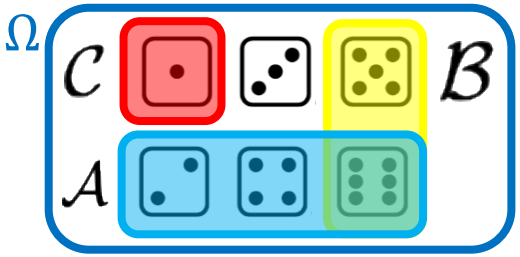
Definition 3.2.2 — Sample Space and Outcomes. The *sample space* Ω is the set of all possible results of the random experiment. An individual result $\omega \in \Omega$ is called an *outcome*. For our robot:



Event

Definition 3.2.2 — Sample Space and Outcomes. The *sample space* Ω is the set of all possible results of the random experiment. An individual result $\omega \in \Omega$ is called an *outcome*. For our robot:

Even Top Face 1



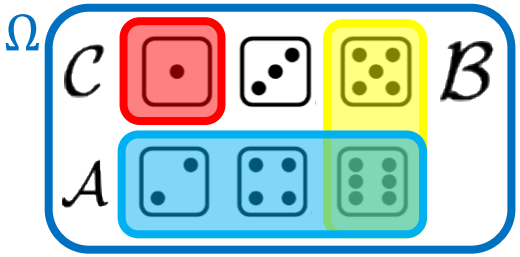
Definition 3.2.3 — Event. An *event* $\mathcal{A}$ is a subset of the sample space Ω (written $\mathcal{A} \subseteq \Omega$). It represents a collection of outcomes to which we can assign a probability. We say an event *occurs* if the outcome ω of the random experiment is an element of $\mathcal{A}$.

- $\mathcal{A} = \{$ ⚀, ⚁, ⚃ $\}$ (The event that the robot is on an **even** face).
- $\mathcal{B} = \{$ ⚄, ⚅ $\}$ (The event that the robot is near the **top** of the die).
- $\mathcal{C} = \{$ ⚀ $\}$ (The elementary event that the robot is on **Face 1**).

Discrete Probability

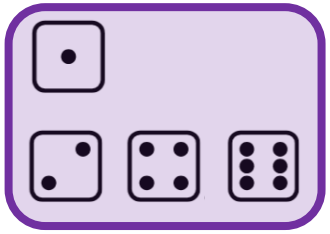
Operations with Events

Even Top Face 1





$\mathcal{A} \cap \mathcal{B}$



$\mathcal{A} \cup \mathcal{B}$



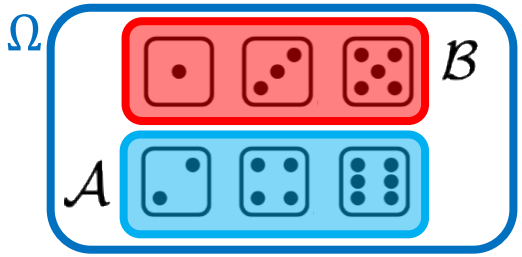
$\mathcal{A}^c$

- $\mathcal{A} = \{$  $,$  $,$  $\}$ (The event that the robot is on an **even** face).
- $\mathcal{B} = \{$  $,$  $\}$ (The event that the robot is near the **top** of the die).
- $\mathcal{C} = \{$  $\}$ (The elementary event that the robot is on **Face 1**).

- **Intersection:** $\mathcal{A} \cap \mathcal{B} = \{$  $\}$. The robot satisfies both conditions only if it is on Face 6.
- **Union:** $\mathcal{A} \cup \mathcal{B} = \{$  $,$  $,$  $,$  $\}$. The robot is either on an even face or a high face.
- **Complement:** $\mathcal{A}^c = \{$ $,$  $,$ $\}$. The robot is not on an even face (it is on an odd face).

Mutually Exclusive Events

Even Odd



Definition 3.2.4 — Mutually Exclusive Event. Two events are *mutually exclusive* (or disjoint) if their intersection is the empty set: $\mathcal{A} \cap \mathcal{B} = \emptyset$. This means it is physically impossible for both to happen at the same time.

Example 3.2.3: Mutually Exclusivity of Odd and Even Faces Events
If $\mathcal{A} = \{$⚁, ⚃, ⚅$\}$ (even faces) and $\mathcal{B} = \{$⚀, ⚂, ⚄$\}$ (odd faces), then $\mathcal{A} \cap \mathcal{B} = \emptyset$. The robot cannot be on an even face and an odd face simultaneously; these events are mutually exclusive.

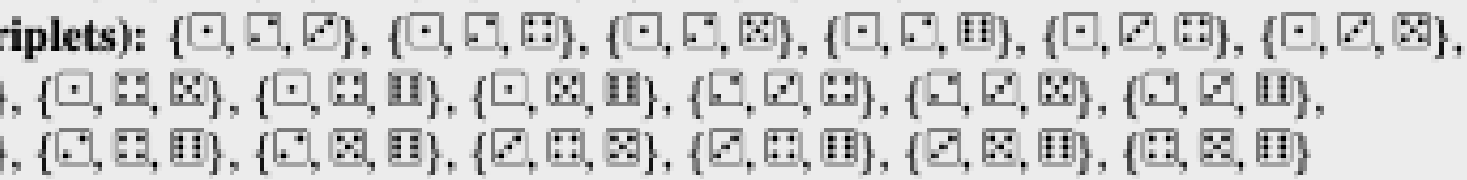
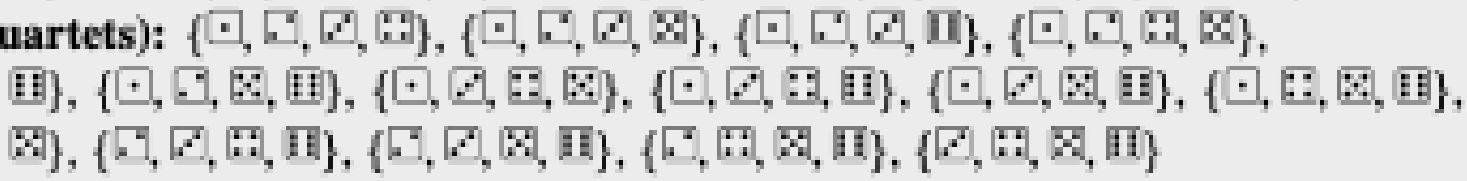
Power Set

Definition 3.2.5 — Power Set.

- 2^{Ω} is the **set of all possible subsets** (events) of the sample space Ω

Example 3.2.4: Power Set of the Robotic on a Die

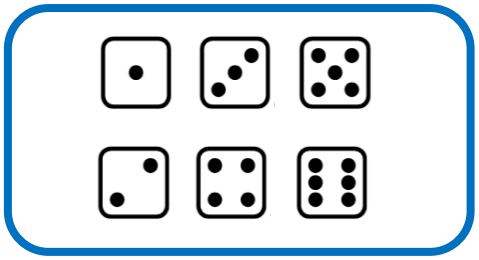
For our robot on a six-sided die ($\Omega = \{1, \ldots, 6\}$), the power set 2^{Ω} contains $2^6 = 64$ subsets. This collection includes every possible grouping of faces, which we can categorize by their size:

- **Size 0 (Impossible):** $\emptyset$
- **Size 1 (Elementary):** 
- **Size 2 (Pairs):** 
- **Size 3 (Triplets):** 
- **Size 4 (Quartets):** 
- **Size 5 (Quintets):** 
- **Size 6 (Certain):** 

Sigma Algebra

Definition 3.2.6 — σ-algebra.

- **Any collection $\Sigma \subset 2^{\Omega}$ so that**
  - $\Omega \subset \Sigma$
  - $A \in \Sigma \Rightarrow A^c \in \Sigma$
  - $A_1, A_2, .. \in \Sigma \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \Sigma$

Ω 

2^{Ω}

Size 0 (Impossible): $\emptyset$
Size 1 (Elementary): {⚀}, {⚁}, {⚂}, {⚃}, {⚄}, {⚅}
Size 2 (Pairs): {⚀,⚁}, {⚀,⚂}, {⚀,⚃}, {⚀,⚄}, {⚀,⚅}, {⚁,⚂}, {⚁,⚃}, {⚁,⚄}, {⚁,⚅}, {⚂,⚃}, {⚂,⚄}, {⚂,⚅}, {⚃,⚄}, {⚃,⚅}, {⚄,⚅}
Size 3 (Triplets): {⚀,⚁,⚂}, {⚀,⚁,⚃}, {⚀,⚁,⚄}, {⚀,⚁,⚅}, {⚀,⚂,⚃}, {⚀,⚂,⚄}, {⚀,⚂,⚅}, {⚀,⚃,⚄}, {⚀,⚃,⚅}, {⚀,⚄,⚅}, {⚁,⚂,⚃}, {⚁,⚂,⚄}, {⚁,⚂,⚅}, {⚁,⚃,⚄}, {⚁,⚃,⚅}, {⚁,⚄,⚅}, {⚂,⚃,⚄}, {⚂,⚃,⚅}, {⚂,⚄,⚅}, {⚃,⚄,⚅}
Size 4 (Quartets): {⚀,⚁,⚂,⚃}, {⚀,⚁,⚂,⚄}, {⚀,⚁,⚂,⚅}, {⚀,⚁,⚃,⚄}, {⚀,⚁,⚃,⚅}, {⚀,⚁,⚄,⚅}, {⚀,⚂,⚃,⚄}, {⚀,⚂,⚃,⚅}, {⚀,⚂,⚄,⚅}, {⚀,⚃,⚄,⚅}, {⚁,⚂,⚃,⚄}, {⚁,⚂,⚃,⚅}, {⚁,⚂,⚄,⚅}, {⚁,⚃,⚄,⚅}, {⚂,⚃,⚄,⚅}
Size 5 (Quintets): {⚀,⚁,⚂,⚃,⚄}, {⚀,⚁,⚂,⚃,⚅}, {⚀,⚁,⚂,⚄,⚅}, {⚀,⚁,⚃,⚄,⚅}, {⚀,⚂,⚃,⚄,⚅}, {⚁,⚂,⚃,⚄,⚅}
Size 6 (Certain): {⚀,⚁,⚂,⚃,⚄,⚅}

✓

$\mathcal{F}$ $\emptyset$ **Even** **Odd** Ω

✓

$\mathcal{G}$ $\emptyset$ Ω

✗

Random Variable

Definition 3.2.7 — Random Variable. A *random variable* $\mathbf{X}$ maps these physical outcomes $\omega \in \Omega$ to a mathematical space $\mathcal{X}$. In our example we define $\mathbf{x} = \mathbf{X}(\omega)$ as the number on the face the robot occupies:

$$\mathbf{X}: \Omega \to \{\boxdot, \boxdot, \boxdot, \boxdot, \boxdot, \boxdot\}$$

$$\boxdot \mapsto 1$$

$$\boxdot \mapsto 2$$

$$\boxdot \mapsto 3$$

$$\boxdot \mapsto 4$$

$$\boxdot \mapsto 5$$

$$\boxdot \mapsto 6$$

Paritition of the Sample Space

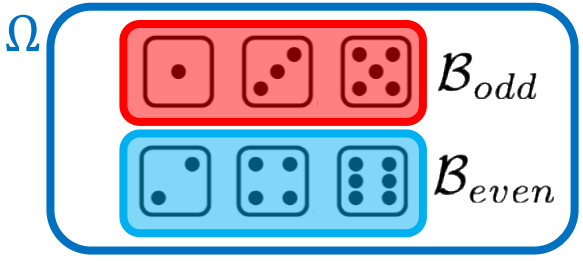
Definition 3.2.8 — Partition of the Sample Space. A collection of events $\{B_1, B_2, \ldots, B_n\}$ is said to be a *partition* of the sample space Ω if it divides the space into distinct, non-overlapping regions that cover every possibility. It must satisfy:

- **Mutually Exclusive**: No two events can occur simultaneously, meaning their intersection is empty.

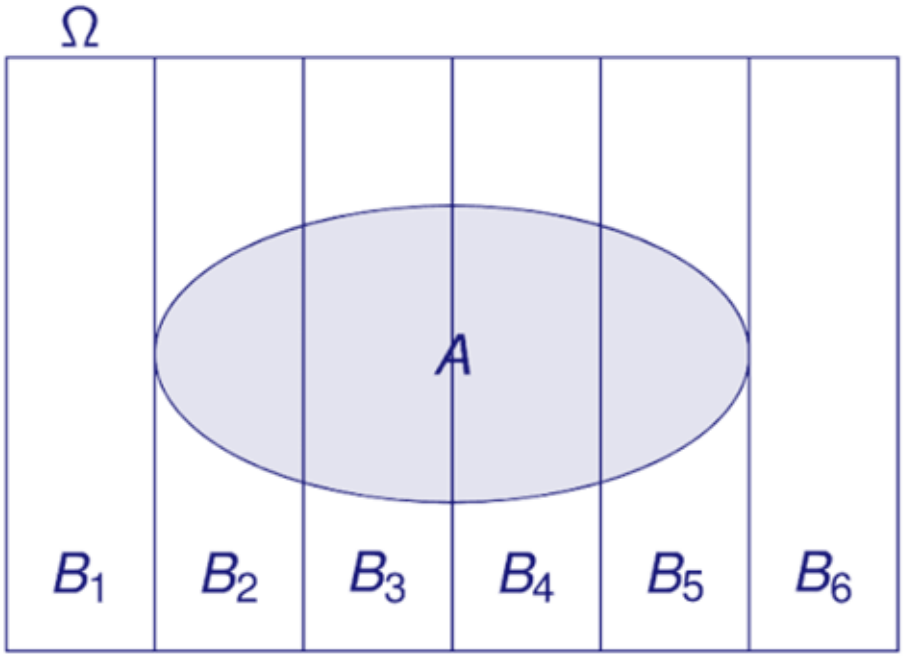
$$B_i \cap B_j = \emptyset, \quad \forall i \neq j$$

- **Collectively Exhaustive**: The union of all events in the collection must equal the entire sample space.

$$\bigcup_{i=1}^{n} B_i = \Omega$$



$\{B_{odd}, B_{Even}\}$ is a partition of Ω



Probability Axioms

Axiom 1 Non-Negativity: $\forall \mathcal{A} \in \Sigma \Rightarrow P(\mathcal{A}) \geq 0.$

$$P(\{\boxdot\}) = P(\{\boxdot\}) = P(\{\boxdot\}) = P(\{\boxdot\}) = P(\{\boxdot\}) = P(\{\boxdot\}) = 1/6 \geq 0$$

$$P(\{\boxdot, \boxdot\}) = P(\{\boxdot, \boxdot\}) = P(\{\boxdot, \boxdot\}) = P(\{\boxdot, \boxdot\}) = P(\{\boxdot, \boxdot\}) = P(\{\boxdot\}) = P(\{\boxdot\}) = P... = 2/6 \geq 0$$

$$P(\{\boxdot, \boxdot, \boxdot\}) = P(\{\boxdot, \boxdot, \boxdot\}) = ... = 3/6 \geq 0$$

$$\dots$$

Axiom 2 Normalization: $P(\Omega) = 1$

$$P(\{\boxdot, \boxdot, \boxdot, \boxdot, \boxdot, \boxdot\}) = 1$$

Axiom 3 Countable Additivity:

$$\text{Let } \mathcal{A}_1, \mathcal{A}_2, \cdots \in \Sigma \ / \ \forall i \neq j, \mathcal{A}_i \bigcap \mathcal{A}_j = \emptyset \quad \Rightarrow P\left(\bigcup_{i=1}^{\infty} \mathcal{A}_i\right) = \sum_{i=1}^{\infty} P(\mathcal{A}_i)$$

$$P(\{\boxdot, \boxdot, \boxdot\}) = P(\{\boxdot\}) + P(\{\boxdot\}) + P(\{\boxdot\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 0.5$$

Properties derived from the Axioms

1. **Probability of the Impossible Event:** The probability of an empty set is zero, $P(\emptyset) = 0$.

$$P(A = \{7\}) = 0$$

2. **Complement Rule:** For any event $\mathcal{A}$, the probability that it does *not* occur is: $P(\mathcal{A}^c) = 1 - P(\mathcal{A})$.

$$P(\boxplus) = \frac{1}{6} \Rightarrow P(\boxdot, \boxdot, \boxdot, \boxplus, \boxplus) = \frac{5}{6}$$

3. **Monotonicity:** If $\mathcal{A} \subseteq \mathcal{B}$, then $P(\mathcal{A}) \leq P(\mathcal{B})$.

$$\frac{1}{6} = P(\boxplus) \leq P(\boxdot, \boxplus, \boxplus) = \frac{1}{2}$$

4. **Numeric Bound:** For any event $\mathcal{A}$, $0 \leq P(\mathcal{A}) \leq 1$.

$$0 \leq P(\{\boxdot, \boxdot, \boxplus\}) = \frac{3}{6} = 0.5 \leq 1$$

Discrete Probability

Probability Space

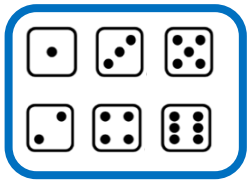
Definition 3.4.1 — Probability Space. A *probability space* is defined by the triplet (Ω, Σ, P), where:

- Ω is the **sample space**, the set of all possible outcomes.
- Σ is the **σ-algebra**, the collection of all events to which the robot can assign a probability.
- P is the **probability measure**, a function that assigns a real number between 0 and 1 to each event in Σ according to the Kolmogorov axioms.

Example 3.4.1: The Robot's Probability Space

For our robot on the six-sided die, we can define the standard probability space $(\Omega, \Sigma_\Omega, P)$ as follows:

- $\Omega = \{$⚀, ⚁, ⚂, ⚃, ⚄, ⚅$\}$.
- $\Sigma_\Omega = 2^\Omega$ (the power set containing all 64 possible subsets of faces).
- $P(A) = \frac{|A|}{6}$ for any event $A \in \Sigma_\Omega$, assuming a fair die.



Ω

Size 0 (Impossible): $\emptyset$
Size 1 (Elementary): {⚀}, {⚁}, {⚂}, {⚃}, {⚄}, {⚅}
Size 2 (Pairs): {⚀,⚁}, {⚀,⚂}, {⚀,⚃}, {⚀,⚄}, {⚀,⚅}, {⚁,⚂}, {⚁,⚃}, {⚁,⚄}, {⚁,⚅}, {⚂,⚃}, {⚂,⚄}, {⚂,⚅}, {⚃,⚄}, {⚃,⚅}, {⚄,⚅}
Size 3 (Triplets): {⚀,⚁,⚂}, {⚀,⚁,⚃}, {⚀,⚁,⚄}, {⚀,⚁,⚅}, {⚀,⚂,⚃}, {⚀,⚂,⚄}, {⚀,⚂,⚅}, {⚀,⚃,⚄}, {⚀,⚃,⚅}, {⚀,⚄,⚅}, {⚁,⚂,⚃}, {⚁,⚂,⚄}, {⚁,⚂,⚅}, {⚁,⚃,⚄}, {⚁,⚃,⚅}, {⚁,⚄,⚅}, {⚂,⚃,⚄}, {⚂,⚃,⚅}, {⚂,⚄,⚅}, {⚃,⚄,⚅}
Size 4 (Quartets): {⚀,⚁,⚂,⚃}, {⚀,⚁,⚂,⚄}, {⚀,⚁,⚂,⚅}, {⚀,⚁,⚃,⚄}, {⚀,⚁,⚃,⚅}, {⚀,⚁,⚄,⚅}, {⚀,⚂,⚃,⚄}, {⚀,⚂,⚃,⚅}, {⚀,⚂,⚄,⚅}, {⚀,⚃,⚄,⚅}, {⚁,⚂,⚃,⚄}, {⚁,⚂,⚃,⚅}, {⚁,⚂,⚄,⚅}, {⚁,⚃,⚄,⚅}, {⚂,⚃,⚄,⚅}
Size 5 (Quintets): {⚀,⚁,⚂,⚃,⚄}, {⚀,⚁,⚂,⚃,⚅}, {⚀,⚁,⚂,⚄,⚅}, {⚀,⚁,⚃,⚄,⚅}, {⚀,⚂,⚃,⚄,⚅}, {⚁,⚂,⚃,⚄,⚅}
Size 6 (Certain): {⚀,⚁,⚂,⚃,⚄,⚅}

Σ_Ω

Probability measure

$\emptyset \to 0$

$\{$⚀$\} \to \frac{1}{6}$ $\{$⚁$\} \to \frac{1}{6}$ $\{$⚂$\} \to \frac{1}{6}$
$\{$⚃$\} \to \frac{1}{6}$ $\{$⚄$\} \to \frac{1}{6}$ $\{$⚅$\} \to \frac{1}{6}$

$\{$⚀,⚁$\} \to \frac{2}{6}$ $\{$⚀,⚂$\} \to \frac{2}{6}$ $\{$⚀,⚃$\} \to \frac{2}{6}$
$\{$⚀,⚄$\} \to \frac{2}{6}$ $\{$⚀,⚅$\} \to \frac{2}{6}$ $\{$⚁,⚂$\} \to \frac{2}{6}$
$\{$⚁,⚃$\} \to \frac{2}{6}$ $\{$⚁,⚄$\} \to \frac{2}{6}$ $\{$⚁,⚅$\} \to \frac{2}{6}$
$\{$⚂,⚃$\} \to \frac{2}{6}$ $\{$⚂,⚄$\} \to \frac{2}{6}$ $\{$⚂,⚅$\} \to \frac{2}{6}$
$\{$⚃,⚄$\} \to \frac{2}{6}$ $\{$⚃,⚅$\} \to \frac{2}{6}$ $\{$⚄,⚅$\} \to \frac{2}{6}$

$\{$⚀,⚁,⚂$\} \to \frac{3}{6}$ $\{$⚀,⚁,⚃$\} \to \frac{3}{6}$ ...

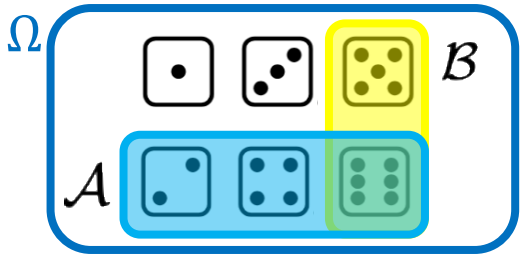
Probability of the Union

The Union (OR Logic)

The *General Addition Rule* calculates the probability that at least one of two events occurs. This is vital when a robot checks if it has entered any one of several regions of interest.

Theorem 3.5.1 — General Addition Rule. For any two events $\mathcal{A}$ and $\mathcal{B}$:

$$P(\mathcal{A} \cup \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) - P(\mathcal{A} \cap \mathcal{B}).$$



Example 3.5.1: Union of Even and High-Face Events

Let $\mathcal{A}$ be the event "even face" $\{2, 4, 6\}$ and $\mathcal{B}$ be the event "face ≥ 5" $\{5, 6\}$. The intersection is $\mathcal{A} \cap \mathcal{B} = \{$⚅$\}$. To find the probability the robot is on an even face *or* a face ≥ 5:

$$P(\mathcal{A} \cup \mathcal{B}) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} \approx 0.67.$$

Without subtracting the intersection, we would "double-count" Face 6, violating the Normalization axiom.

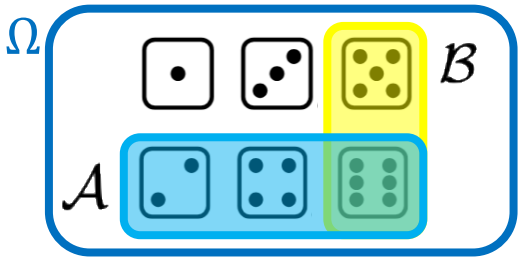
Conditional Probability

Definition 3.5.1 — Conditional Probability. The probability of an event $\mathcal{A}$ occurring, given that event $\mathcal{B}$ has already occurred, is:

$$P(\mathcal{A} \mid \mathcal{B}) = \frac{P(\mathcal{A} \cap \mathcal{B})}{P(\mathcal{B})}, \quad \text{for } P(\mathcal{B}) > 0.$$

Example 3.5.2: Probability of Even-Face Knowing the Robot Is on a High-Face
Suppose we know for certain the robot is on a face ≥ 5 (event $\mathcal{B}$). What is the probability it is on an even face (event $\mathcal{A}$)? Our sample space effectively shrinks to $\{5, 6\}$. Since only Face 6 is even:

$$P(\text{even} \mid \text{face} \geq 5) = \frac{P(\{\boxdot, \boxdot\!\boxdot, \boxdot\!\boxdot\!\boxdot\} \cap \{\boxtimes, \boxplus\})}{P(\{\boxtimes, \boxplus\})} = \frac{P(\{\boxplus\})}{P(\{\boxtimes, \boxplus\})} = \frac{1/6}{2/6} = 0.5.$$



Intersection of Events

Theorem 3.5.2 — The Product Rule. The probability of the intersection of two events is:

$$P(A \cap B) = P(A \mid B)P(B).$$

Example 3.5.3: Probability of Face-2 Event and Even-Face Event

Let A be the event "the robot is on **Face 2**" and B be the event "the sensor reports **even**." If we have a uniform prior $P(A) = 1/6$ and a sensor model where $P(B \mid A) = 0.9$, the joint probability of being on Face 2 *and* the sensor correctly reporting even is:

$$P(\{\square\} \cap \{\square, \square, \square\}) = P(\{\square, \square, \square\} \mid \{\square\})P(\{\square\}) = 0.9 \cdot \frac{1}{6} = 0.15.$$

Independent Events

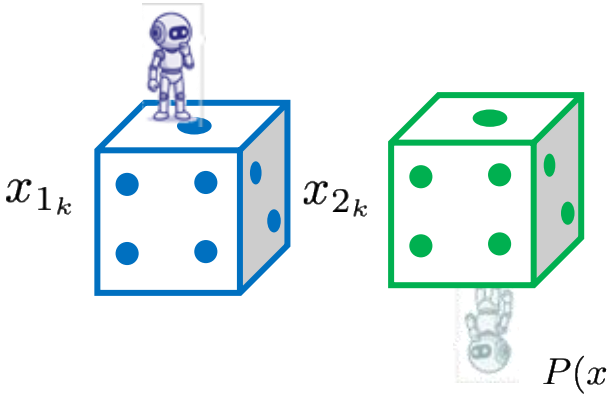
Definition 3.5.2 — Independence. Two events $\mathcal{A}$ and $\mathcal{B}$ are independent if and only if:

$$P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A})P(\mathcal{B}).$$

Example 3.5.4: Two Robots on Their Dice

If we have two robots on two different dice, the event "Robot 1 is on Face 1" and "Robot 2 is on Face 6" are independent. However, for a single robot, the event "Located on Face 2" and "Sensor detects even" are **dependent**, as the sensor reading is physically caused by the position.

$$\boldsymbol{x}_k = [x_{1_k}\ x_{2_k}]$$



$$\Omega = \left\{ \begin{array}{cccccc} (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \\ (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \\ (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \\ (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \\ (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \\ (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) & (\boxdot,\boxdot) \end{array} \right\}.$$

$$P(x_{1_k} = \boxdot) = 1/6$$
$$P(x_{2_k} = \boxdot) = 1/6$$
$$P(x_{1_k} = \boxdot \cap x_{2_k} = \boxdot) = P(x_{1_k} = \boxdot \cdot P(x_{2_k} = \boxdot) = 1/6 * 1/6 = 1/36$$
$$P(x_{1_k} = \boxdot \cap x_{2_k} = \boxdot) = P(x_k = (\boxdot, \boxdot)) = 1/36$$

Independent Events

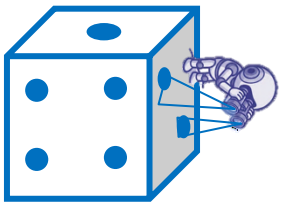
Independent Events

Definition 3.5.2 — Independence. Two events $\mathcal{A}$ and $\mathcal{B}$ are independent if and only if:

$$P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A})P(\mathcal{B}).$$

Example 3.5.4: Two Robots on Their Dice
If we have two robots on two different dice, the event "Robot 1 is on Face 1" and "Robot 2 is on Face 6" are independent. However, for a single robot, the event "Located on Face 2" and "Sensor detects even" are **dependent**, as the sensor reading is physically caused by the position.

Dependent Events





$$P(z_k = Even)$$
$$P(x_k = \boxdot)$$

$$\left.\begin{array}{c} P(z_k = Even | x_k = 2) \neq P(z_k = Event) \\ 0.9 \neq 0.5 \end{array}\right\} \Rightarrow z_k \text{ does depends on } x_k$$

Conditional Independent Events

Conditional Independence

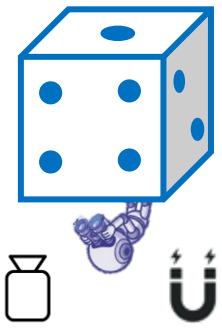
While two events might be dependent in general, they can become independent if we know the value of a third, intervening variable. This is known as *conditional independence*.

Definition 3.5.3 — Conditional Independence. Two events $\mathcal{A}$ and $\mathcal{B}$ are conditionally independent given $\mathcal{C}$ if and only if:

$$P(\mathcal{A} \cap \mathcal{B} \mid \mathcal{C}) = P(\mathcal{A} \mid \mathcal{C})P(\mathcal{B} \mid \mathcal{C}).$$

Equivalently, this implies that if $\mathcal{C}$ is known, then the occurrence of $\mathcal{B}$ provides no additional information about the likelihood of $\mathcal{A}$:

$$P(\mathcal{A} \mid \mathcal{B}, \mathcal{C}) = P(\mathcal{A} \mid \mathcal{C}).$$



$$P(z_{c_k} = \boxdot)$$
$$P(z_{m_k} = \boxdot)$$

Dependent Events if we don't know where the robot is

$$P(z_{c_k} = \boxdot \mid x_k = \boxdot)$$
$$P(z_{m_k} = \boxdot \mid x_k = \boxdot)$$

Independent Events if we know the robot is on face 6

PMF

Definition 3.6.1 — Probability Mass Function. For a discrete random variable $\mathbf{x}$, the *probability mass function* (PMF), denoted as $p(x)$, assigns a probability to each possible realization $x \in \mathcal{X}$. The PMF must satisfy two conditions:

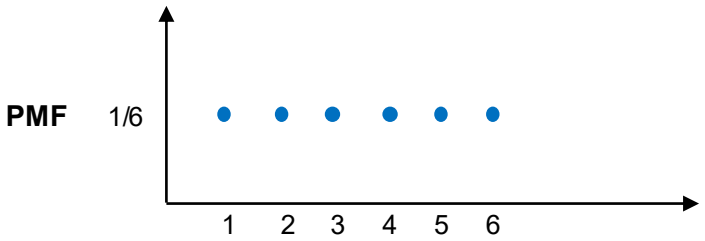
1. $p(x) \geq 0$ for all $x \in \mathcal{X}$.
2. $\sum_{x \in \mathcal{X}} p(x) = 1$.

Example 3.6.1: Uniform PMF for the Robot on a Die

If the robot is placed on the die by a fair process, we describe its initial state with a *uniform distribution*. Since there are six possible faces in our sample space $\mathcal{X}$, the probability is spread equally:

$$p(x) = \frac{1}{6}, \quad \text{for } x = 1, \ldots, 6.$$

Summing these probabilities yields $\sum_{i=1}^{6} \frac{1}{6} = 1$, which satisfies the normalization requirement of the second axiom.



Joint PMF

Definition 3.6.2 — Joint PMF. The *joint probability mass function* of two discrete random variables $\mathbf{x}$ and $\mathbf{z}$ is a function $p(x, z)$ that assigns a probability to the intersection of the events $\{\mathbf{x} = x\}$ and $\{\mathbf{z} = z\}$:

$$p(x, z) = P(\mathbf{x} = x \cap \mathbf{z} = z)$$

The joint PMF must satisfy:

1. $p(x, z) \geq 0$ for all pairs (x, z).
2. $\sum_{x \in \mathcal{X}} \sum_{z \in \mathcal{Z}} p(x, z) = 1$.



$$P(x = 4 \cap z = Even)$$
$$P(x = 4, z = Even)$$

Example

$p(x, z)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$	$x = 6$
$z = 0$ (Even)	0.017	0.150	0.016	0.150	0.017	0.150
$z = 1$ (Odd)	0.150	0.017	0.150	0.016	0.150	0.016

$$P(x = 4, z = Odd)$$
$$P(x = 4 \cap z = Odd)$$

Marginalization

Marginalization: Recovering Individual Probabilities

If we have the joint distribution $p(x, z)$, we can recover the distribution of a single variable by summing over all possible values of the other. This process is called **marginalization**.

Theorem 3.6.1 — Marginal PMF. The *marginal probability mass function* of $\mathbf{x}$ is obtained by summing the joint PMF over all possible values of z:

$$p(x) = \sum_{z \in \mathcal{Z}} p(x, z)$$

Similarly, the marginal PMF of $\mathbf{z}$ is:

$$p(z) = \sum_{x \in \mathcal{X}} p(x, z)$$

Example

$p(x, z)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$	$x = 6$	$p(z)$
$z = 0$ (Even)	0.017	0.150	0.016	0.150	0.017	0.150	1/2
$z = 1$ (Odd)	0.150	0.017	0.150	0.016	0.150	0.016	1/2
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	

$\mathbf{p(z = 0) = \Sigma_i p(x_i, z = 0)}$

$\mathbf{p(x = 4) = p(x = 4, z = 0) + p(x = 4, z = 1)}$

Discrete Probability

Sensor Model

The *conditional probability* $p(z \mid x)$ represents the likelihood of observing a specific sensor measurement z given that the robot is in a known state x. In robotics, this is often referred to as the **sensor model** or **observation model**. It describes the reliability and noise characteristics of the hardware.

Example: Parity Sensor Model



$$\begin{cases} P(z_k = Odd \mid x_k = Odd) = \dfrac{\overbrace{p(z_k = Odd, x_k = Odd)}^{3*0.150}}{\underbrace{p(x_k = Odd)}_{0.5}} = 0.9 \\ P(z_k = Even \mid x_k = Odd) = 0.1 \end{cases}$$

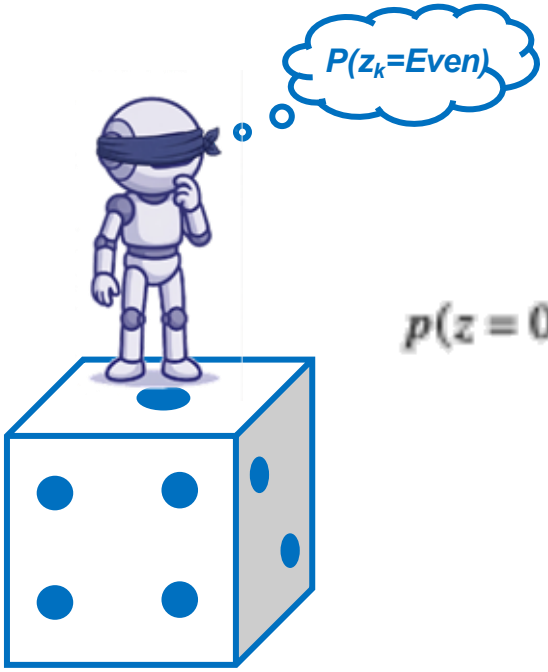
$$\begin{cases} P(z_k = Odd \mid x_k = Even) = 0.1 \\ P(z_k = Even \mid x_k = Odd) = 0.9 \end{cases}$$

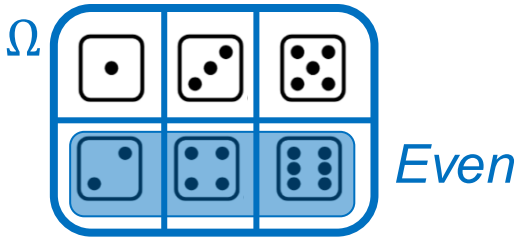
$p(x, z)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$	$x = 6$
$z = 0$ (Even)	0.017	0.150	0.016	0.150	0.017	0.150
$z = 1$ (Odd)	0.150	0.017	0.150	0.016	0.150	0.016

The Law of Total Probability

Theorem 3.7.2 — Law of Total Probability. Let X be a discrete random variable with support $\mathcal{X}$. If the events $\{X = x\}$, for $x \in \mathcal{X}$, form a partition of the sample space, the probability mass function of Z satisfies

$$p(z) = \sum_{x \in \mathcal{X}} p(z \mid x)\, p(x). \tag{3.10}$$





Prediction of Even without Observing

$$p(z=0) = \sum_{x=1}^{6} p(z=0 \mid x)p(x)$$

$$= [p(0|1) + p(0|2) + p(0|3) + p(0|4) + p(0|5) + p(0|6)] \cdot \frac{1}{6}$$

$$= [0.1 + 0.9 + 0.1 + 0.9 + 0.1 + 0.9] \cdot \frac{1}{6}$$

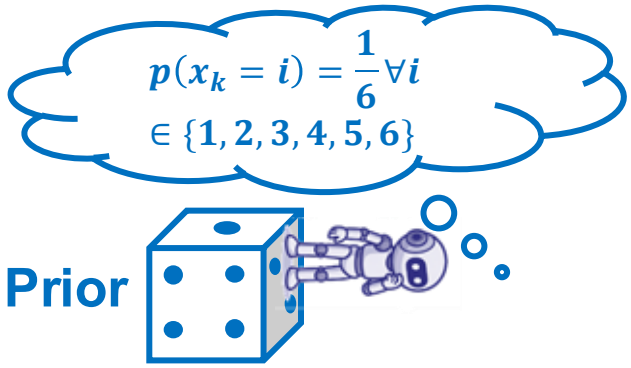
$$= 3.0 \cdot \frac{1}{6} = 0.5$$

Bayes Rule

Theorem 3.7.3 — Bayes' Rule. For two discrete random variables **x** and **z**:

$$p(x \mid z) = \frac{p(z \mid x)p(x)}{p(z)}$$

where $p(x \mid z)$ is the **posterior**, $p(z \mid x)$ is the **likelihood** (sensor model), $p(x)$ is the **prior**, and $p(z)$ is the **evidence** (calculated via the Law of Total Probability).



1. **Prior:** $p(x = 2) = 0.167$

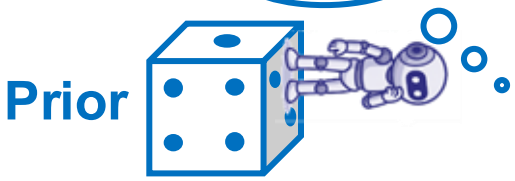
Bayes Rule

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$$p(x \mid z) = \frac{p(z \mid x)p(x)}{p(z)}$$

where $p(x \mid z)$ is the **posterior**, $p(z \mid x)$ is the **likelihood** (sensor model), $p(x)$ is the **prior**, and $p(z)$ is the **evidence** (calculated via the Law of Total Probability).

$p(x_k = i) = \frac{1}{6} \forall i \in \{1, 2, 3, 4, 5, 6\}$



$z_k = Even$



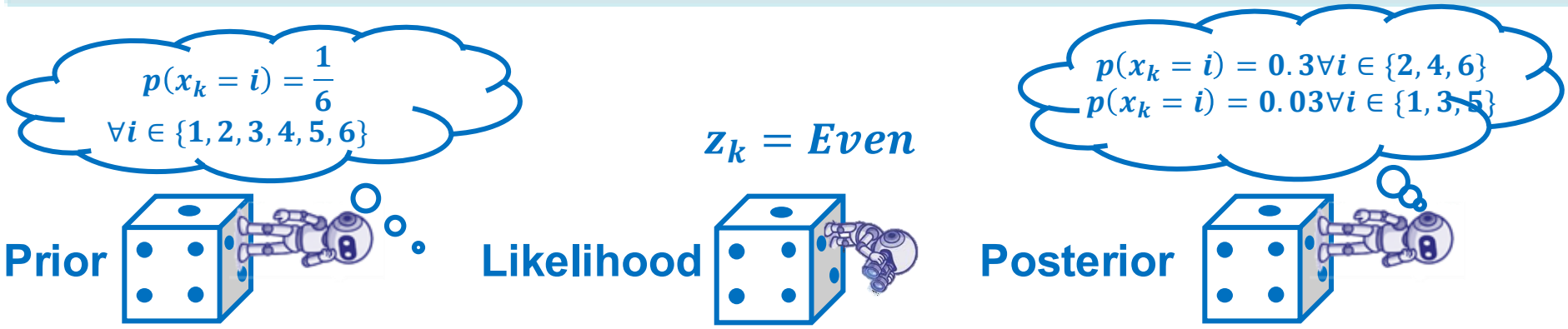
1. **Prior:** $p(x = 2) = 0.167$
2. **Likelihood:** $p(z = 0 \mid x = 2) = 0.9$ (The sensor is 90% accurate for even faces).

Bayes Rule

Theorem 3.7.3 — Bayes' Rule. For two discrete random variables **x** and **z**:

$$p(x \mid z) = \frac{p(z \mid x)p(x)}{p(z)}$$

where $p(x \mid z)$ is the **posterior**, $p(z \mid x)$ is the **likelihood** (sensor model), $p(x)$ is the **prior**, and $p(z)$ is the **evidence** (calculated via the Law of Total Probability).



1. **Prior:** $p(x = 2) = 0.167$
2. **Likelihood:** $p(z = 0 \mid x = 2) = 0.9$ (The sensor is 90% accurate for even faces).
3. **Evidence:** $p(z = 0) = 0.5$ (from our previous calculation).
4. **Posterior:**

$$p(x = 2 \mid z = 0) = \frac{0.9 \cdot 0.167}{0.5} = 0.3006$$

Sequence Joint Probability

$$
\begin{aligned}
p(\mathbf{x}_0) & \\
p(\mathbf{x}_0, \mathbf{x}_1) &= p(\mathbf{x}_0) p(\mathbf{x}_1 \mid \mathbf{x}_0) \\
p(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2) &= \underbrace{p(\mathbf{x}_0) p(\mathbf{x}_1 \mid \mathbf{x}_0)}_{p(\mathbf{x}_0, \mathbf{x}_1)} p(\mathbf{x}_2 \mid \mathbf{x}_0 \mathbf{x}_1) \\
& \vdots \\
p(\mathbf{x}_0, \ldots, \mathbf{x}_k) &= p(\mathbf{x}_0) \prod_{i=1}^{k} p(\mathbf{x}_i \mid \mathbf{x}_0, \ldots, \mathbf{x}_{i-1})
\end{aligned}
$$

Stochastic Process

Definition 3.8.1 — Stochastic Process. A *stochastic process* is a collection of random variables $\{\mathbf{x}_t\}_{t \in T}$ defined on the same probability space (Ω, Σ, P) and indexed by a set T. In a general discrete-time process, the probability of being in a certain state at time k may depend on the **entire history** of the process:

$$p(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{x}_{k-2}, \ldots, \mathbf{x}_0) \tag{3.11}$$

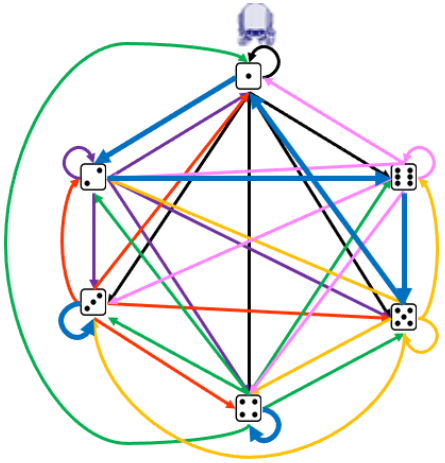
In our die-world, we focus on **discrete-time processes**, where the index set $T = \{0, 1, 2, \ldots\}$ represents sequential time steps (e.g., before and after movement commands).

- **State Sequence:** The sequence $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k$ represents the robot's trajectory across the faces.
- **Realization:** While the process is a collection of probabilities, a specific path taken by the robot (e.g., $1 \to 2 \to 3 \to 3$) is called a *realization* or *sample path* of the process.

Stochastic Process Example

Battery decay via motion history

$$p(x_{k+1} = j \mid x_{0:k}) = \begin{cases} \gamma^{N_k} & j \in \mathrm{adj}(x_k), \\ 1 - \gamma^{N_k}, & j = x_k, \\ 0, & \text{otherwise}, \end{cases}$$



adj()

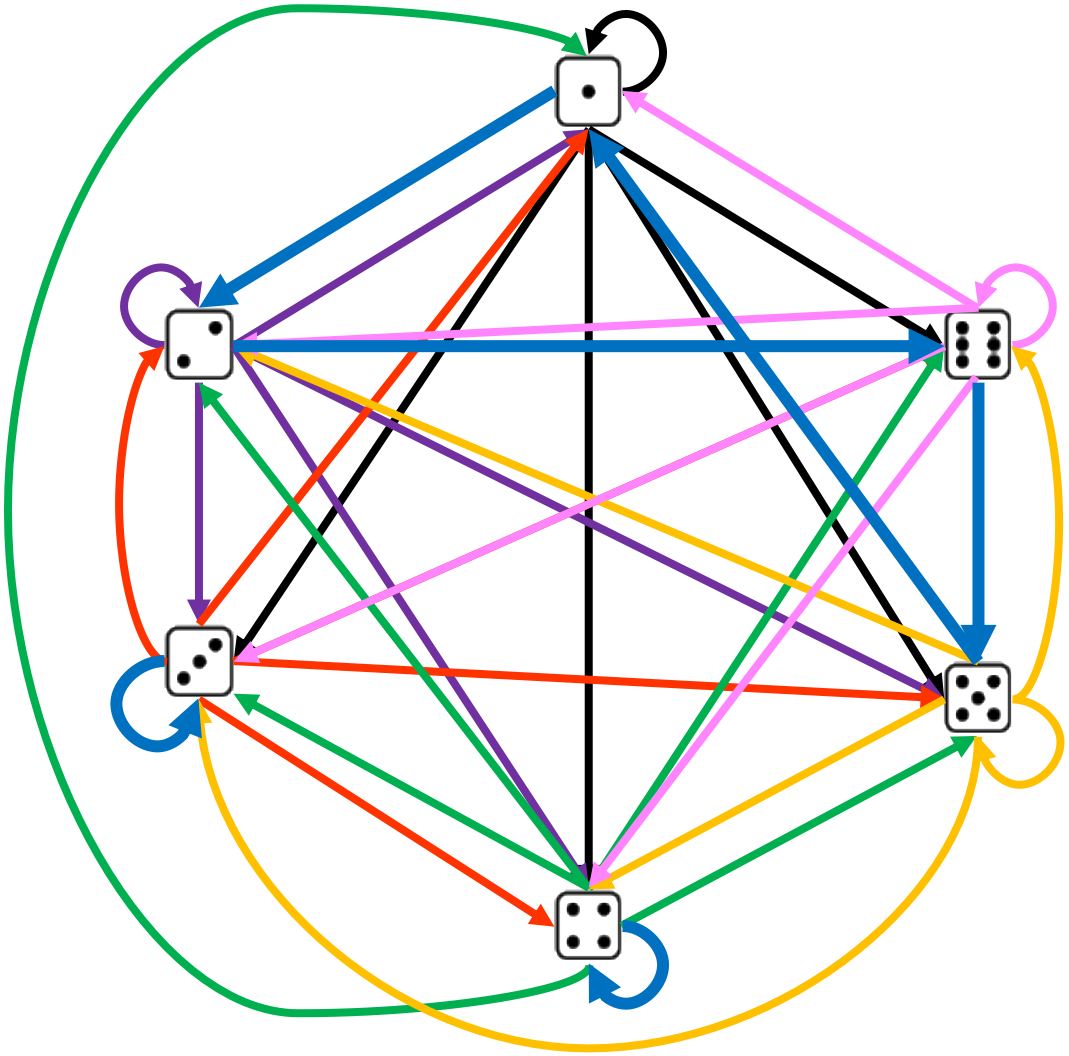
x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4

$\boldsymbol{\lambda = 0.5}$

- To simplify, let us focus on **F motion only** commands
- **N_k:** number of successfuk **motion steps** so far
- The probability of **succesfully reaching** the new state **decreases with N_k** because the **battery decays**

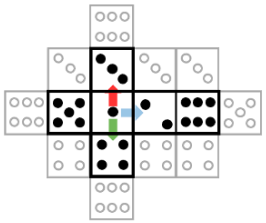
Discrete Probability

Forward Motion Model



adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4



Successful F motion 0.5^{N_k}

Failure F motion from

Failure F motion from

Failure F motion from

Failure F motion from

Failure F motion from

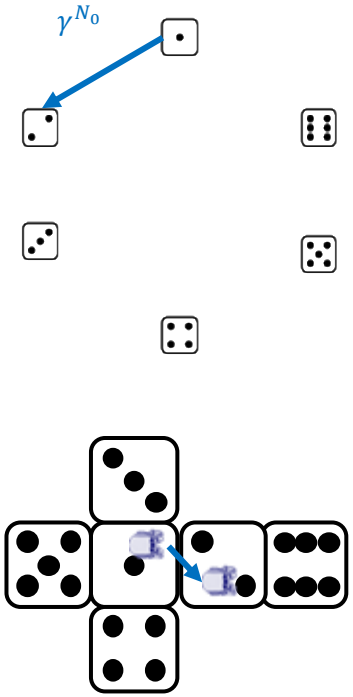
Failure F motion from

$\frac{1-0.5^{N_k}}{5}$

N_k: number of accumulated succesful transitions

Discrete Probability

1st Sequence example

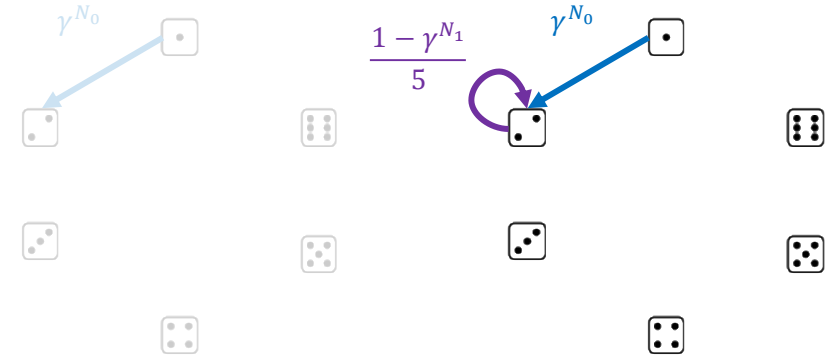


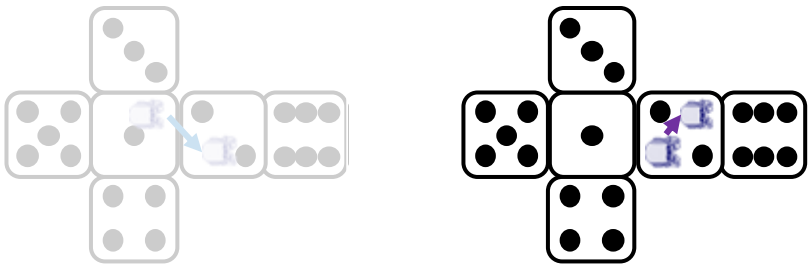
K=0
N_0=1

$$p(x_1 = 2|1) = \gamma^{N_0} = 0.5$$

Discrete Probability

1st Sequence example





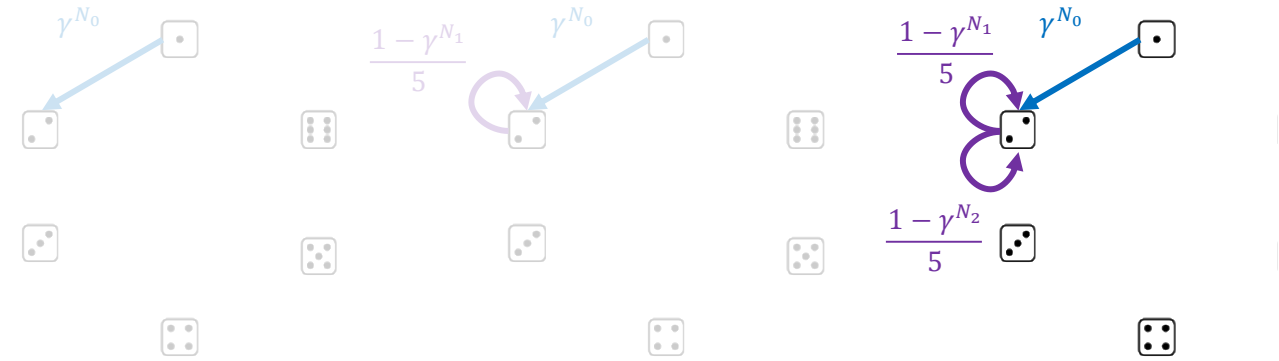
K=0
N_0=1

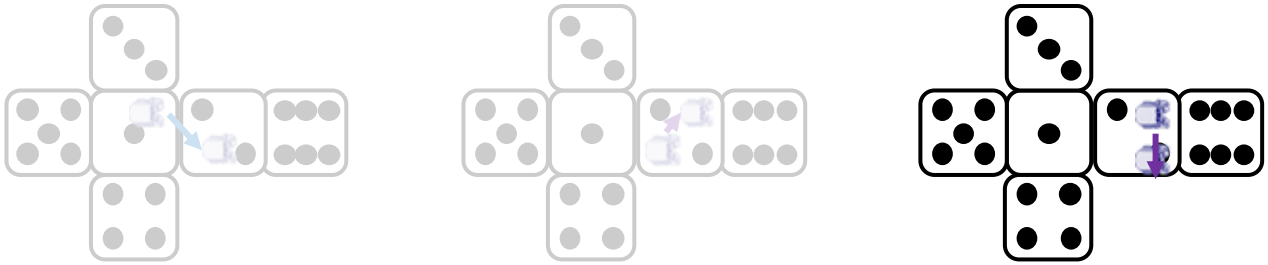
K=1
N_1=1

$$p(x_2 = 2|1,2) = p(x_2 = 2|2) \cdot p(x_1 = 2|1) = \frac{(1-\gamma^{N_1})}{5} \cdot 0.5 = 0.1 \cdot 0.5 = 0.05$$

Discrete Probability

1st Sequence example





K=0
N_0=1

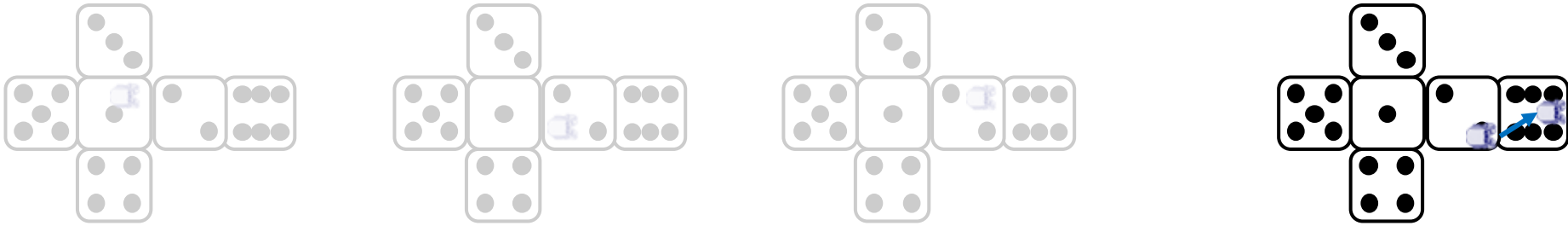
K=1
N_1=1

K=2
N_2=1

$$p(x_3 = 2|1,2,2) = p(x_2 = 2|2) \cdot p(x_2 = 2|1,2) = \frac{(1-\gamma^{N_2})}{5} \cdot 0.05 = 0.1 \cdot 0.05 = 0.005$$

Discrete Probability

1st Sequence example



K=0
N_0=1

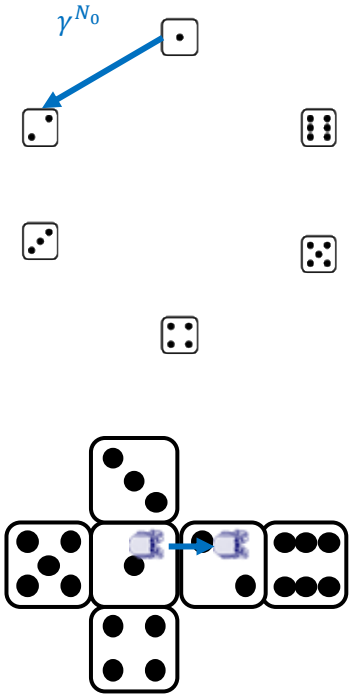
K=1
N_1=1

K=2
N_2=1

K=3
N_3=2

$$p(x_4 = 6|1,2,2,2) = p(x_3 = 6|2) \cdot p(x_3 = 2|1,2,2) = \gamma^{N_3} \cdot 0.005 = 0.5^2 \cdot 0.005 = 0.00125$$

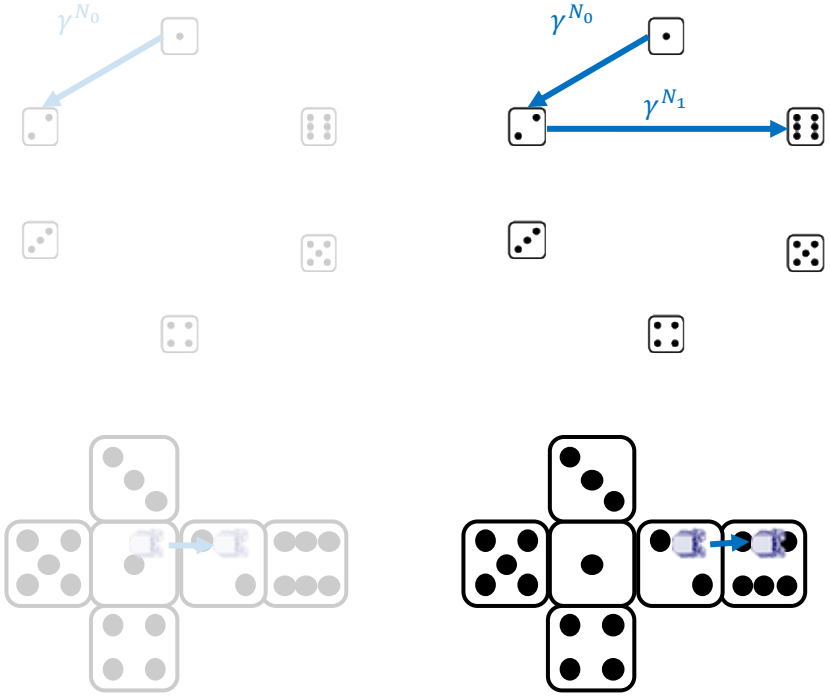
2nd Sequence example



K=0
N_0=1

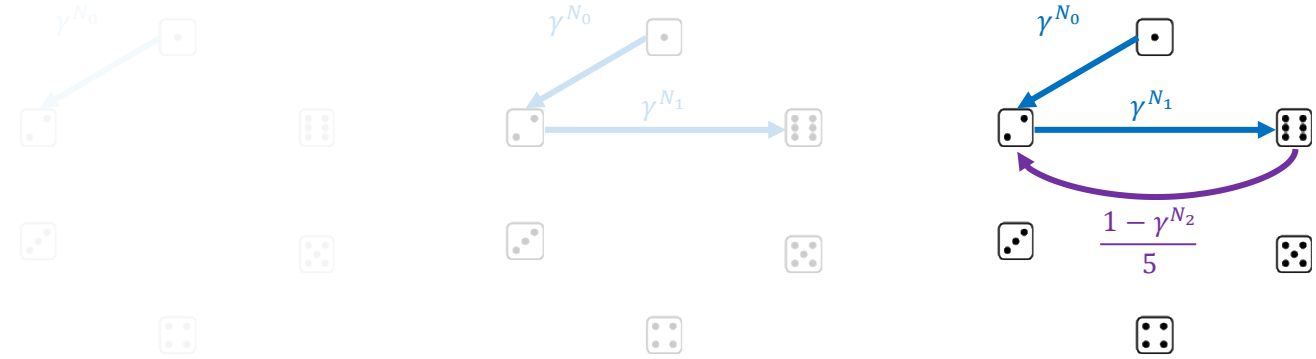
$$p(x_1 = 2|1) = \gamma^{N_0} = 0.5$$

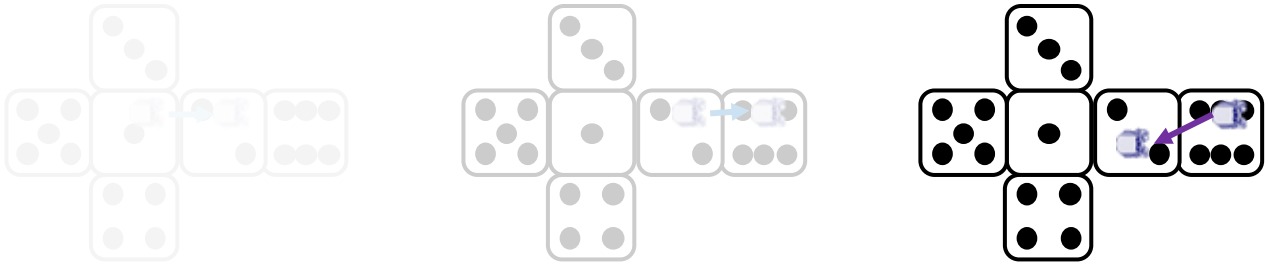
2nd Sequence example



$$p(x_2 = 6|1,2) = p(x_2 = 6|2) \cdot p(x_1 = 2|1) = \gamma^{N_1} \cdot 0.5 = 0.25 \cdot 0.5 = 0.125$$

2nd Sequence example





K=0
N_0=1

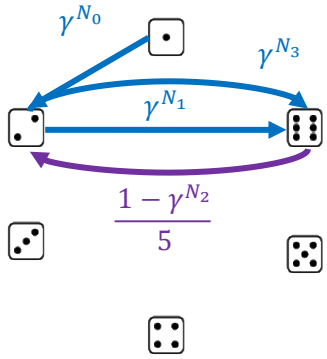
K=1
N_1=2

K=2
N_2=2

$$p(x_3 = 2|1,2,6) = p(x_2 = 2|6) \cdot p(x_1 = 6|1,2) = \frac{(1-\gamma^{N_2})}{5} \cdot 0.125 = 0.15 \cdot 0.125 = 0{,}01875$$

Discrete Probability

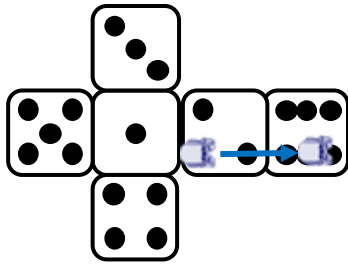
2nd Sequence example



K=0
N_0=1

K=1
N_1=2

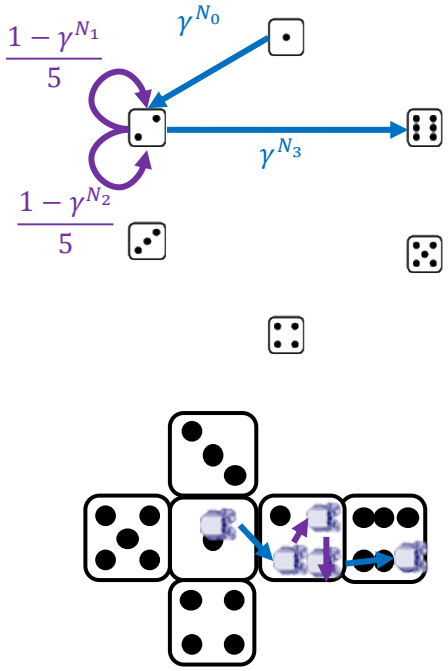
K=2
N_2=2



K=3
N_3=3

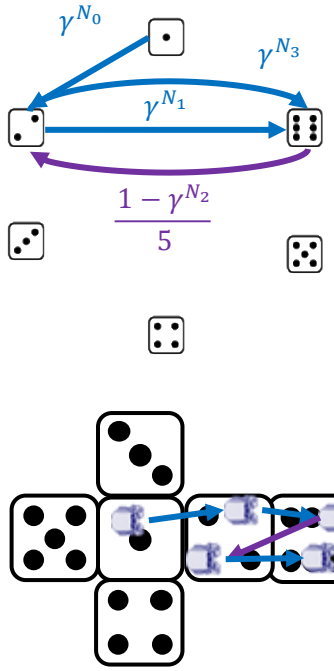
$$p(x_4 = 6|1,2,6,2) = p(x_2 = 2|2) \cdot p(x_1 = 2|1,2,6) = \gamma^{N_3} \cdot 0.01875 = 0.125 \cdot 0.01875 = 0.00234375$$

Discrete Probability



K=3
N_3=2

The **probability** of reaching **6 from 2 depends** on the trajectory **history**



K=3
N_3=2

$$\mathbf{p(x_4 = 6|1,2,6,2)} = \frac{\boldsymbol{p(1,2,6,2,6)}}{\boldsymbol{p(1,2,6,2)}} = \frac{0.00234375}{0{,}01875} = \mathbf{0.125}$$

$$\mathbf{p(x_4 = 6|1,2,2,2)} = \frac{\boldsymbol{p(1,2,2,2,6)}}{\boldsymbol{p(1,2,2,2)}} = \frac{0.00125}{0{,}005} = \mathbf{0.25}$$

Markovian Process

Definition 3.8.2 Markov Property A stochastic process $\{\boldsymbol{x}_k\}$ satisfies the **Markov property** if the conditional probability of the future state $\boldsymbol{x}_k$ depends only on the current state $\boldsymbol{x}_{k-1}$, and is independent of all previous states:

$$p(\boldsymbol{x}_k \mid \boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{k-1}) = p(\boldsymbol{x}_k \mid \boldsymbol{x}_{k-1})$$

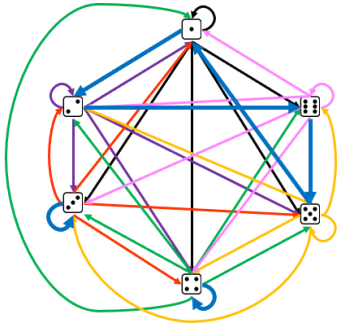
$\mathbf{p(x_4 = 6|2, 2, 2, 1)} \neq \mathbf{p(x_4 = 6|2, 6, 2, 1)} \neq \mathbf{p(x_4 = 6|2)} \Rightarrow$ **Non Markovian!!**

Markovian Process

Definition 3.8.2 Markov Property A stochastic process $\{\boldsymbol{x}_k\}$ satisfies the **Markov property** if the conditional probability of the future state $\boldsymbol{x}_k$ depends only on the current state $\boldsymbol{x}_{k-1}$, and is independent of all previous states:

$$p(\boldsymbol{x}_k \mid \boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{k-1}) = p(\boldsymbol{x}_k \mid \boldsymbol{x}_{k-1})$$

$$p(x_k = j | u_k = F, x_{k-1} = i) = \begin{cases} 0.5 & \text{if } j \in \text{adj(F, i)} \\ 0.1 & \text{otherwise} \end{cases}$$



adj()

x_{k-1}	L	F	R
1	3	2	4
2	3	6	4
3	6	3	1
4	1	4	6
5	3	1	4
6	3	5	4

Successful transitions appear in blue

$$\mathbf{p(x_4 = 6|1, 2, 6, 2)} = \frac{\boldsymbol{p(1, 2, 6, 2, 6)}}{\boldsymbol{p(1, 2, 6, 2)}} = \frac{\mathbf{1 \cdot 0.5 \cdot 0.5 \cdot 0.1 \cdot 0.5 \cdot}}{\mathbf{1 \cdot 0.5 \cdot 0.5 \cdot 0.1}} = \mathbf{0.5}$$

$$\mathbf{p(x_4 = 6|1, 2, 2, 2)} = \frac{\boldsymbol{p(1, 2, 2, 2, 6)}}{\boldsymbol{p(1, 2, 2, 2)}} = \frac{\mathbf{1 \cdot 0.5 \cdot 0.12 \cdot 0.5 \cdot}}{\mathbf{1 \cdot 0.5 \cdot 0.12}} = \mathbf{0.5}$$

Markovian!!

Definition 3.8.2 Markov Property A stochastic process $\{\boldsymbol{x}_k\}$ satisfies the **Markov property** if the conditional probability of the future state $\boldsymbol{x}_k$ depends only on the current state $\boldsymbol{x}_{k-1}$, and is independent of all previous states:

$$p(\boldsymbol{x}_k \mid \boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{k-1}) = p(\boldsymbol{x}_k \mid \boldsymbol{x}_{k-1})$$

$$p(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k) = p(\mathbf{x}_0) \prod_{i=1}^{k} p(\mathbf{x}_i \mid \mathbf{x}_0, \ldots, \mathbf{x}_{i-1})$$

$\Downarrow$ If Markovian Simplifies to…

$$p(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k) = p(\mathbf{x}_0) \prod_{i=1}^{k} p(\mathbf{x}_i \mid \mathbf{x}_{i-1})$$

Markov Chain

Definition 1.8.3 — Markov Chain. Markov Chain A discrete-time Markov chain is defined by the triplet $\{\mathcal{X}, \mathbf{T}, \pi_0\}$, where:

$\mathcal{X} = \{1, \ldots, n\}$ is the **state space**, containing all possible states.

- $\mathbf{T}$ is the **transition matrix**. Adopting the *column-vector belief convention*, each matrix element is defined as

$$t_{ij} = p(x_k = i \mid x_{k-1} = j),$$

where each column contains the conditional probability distribution of the next state given the current state:

$$\mathbf{T} = \begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1n} \\ t_{21} & t_{22} & \cdots & t_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ t_{n1} & t_{n2} & \cdots & t_{nn} \end{bmatrix} = \begin{bmatrix} p(x_k = 1 \mid x_{k-1} = 1) & p(x_k = 1 \mid x_{k-1} = 2) & \cdots & p(x_k = 1 \mid x_{k-1} = n) \\ p(x_k = 2 \mid x_{k-1} = 1) & p(x_k = 2 \mid x_{k-1} = 2) & \cdots & p(x_k = 2 \mid x_{k-1} = n) \\ \vdots & \vdots & \ddots & \vdots \\ p(x_k = n \mid x_{k-1} = 1) & p(x_k = n \mid x_{k-1} = 2) & \cdots & p(x_k = n \mid x_{k-1} = n) \end{bmatrix}.$$

Since each column is a probability mass function,

$$t_{ij} \geq 0, \qquad \sum_{i=1}^{n} t_{ij} = 1, \quad \forall j \in \mathcal{X}.$$

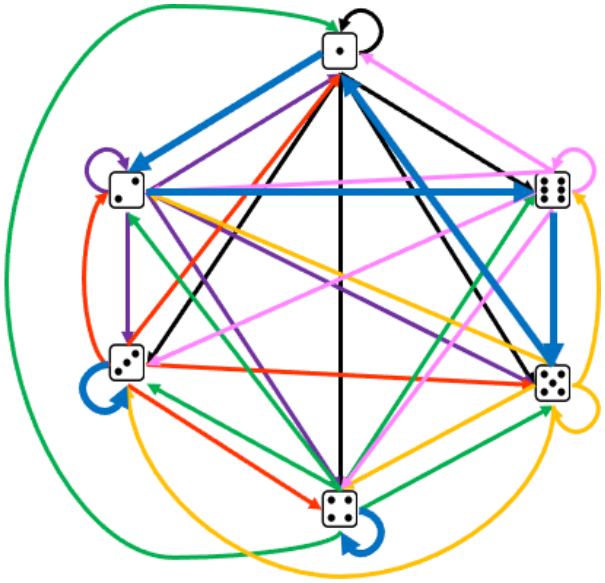
Initial Belief: $\pi_0 = \begin{bmatrix} p(x_0 = 1) \\ p(x_0 = 2) \\ \vdots \\ p(x_0 = n) \end{bmatrix}.$

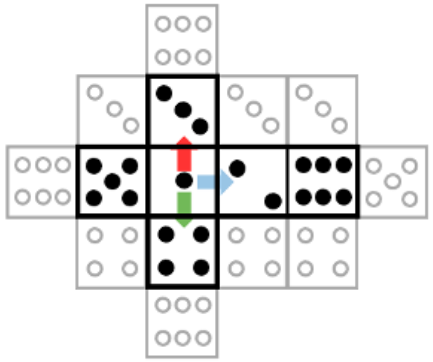
Example: Forward Motion Command only

State Space: $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$

State Transition Matrix: $\mathbf{T}_F =$

$$\begin{array}{c} \\ x_k \end{array} \overset{x_{k-1}}{\begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix}}.$$





Initial Belief: $\mathbf{p_0} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$

Hidden MarKov Mocel (HMM)

Definition 3.8.4 — Hidden Markov Model. Hidden Markov Model A discrete HMM is formally defined by the quintuple $\{\mathcal{X}, \mathcal{Z}, \mathbf{T}, \mathbf{O}, \pi\}$, where:

$\mathcal{X}$ is the set of possible states (e.g., the faces $\{1, \ldots, 6\}$).

- $\mathcal{Z}$ is the set of possible observations (e.g., the parity sensor readings $\{0, 1\}$).
- $\mathbf{T}$ is the **Transition Model**, $p(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{u}_k)$, representing the dynamics under control $\mathbf{u}$.
- $\mathbf{O}$ is the **Observation Model**, $p(\mathbf{z}_k \mid \mathbf{x}_k)$, mapping the hidden state to the perceived evidence.
- π is the **Initial Belief**, representing the starting probability distribution $\mathbf{p}_0$.

Example: Forward Motion Command only

State Space: $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$

State Transition Matrix: $\mathbf{T}_F = \begin{matrix} \\ x_k \end{matrix} \overset{x_{k-1}}{\begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix}}$.

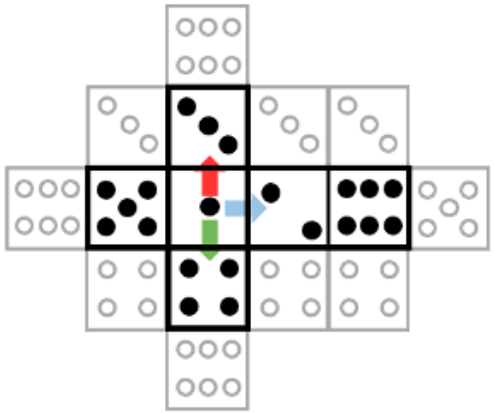
Initial Belief at k-1: $\mathbf{p_{k-1}} = \begin{bmatrix} 0.4 & 0.2 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix}^T$

Observation Model: $\mathbf{O} = \begin{bmatrix} 0.9 & 0.1 & 0.9 & 0.1 & 0.9 & 0.1 \\ 0.1 & 0.9 & 0.1 & 0.9 & 0.1 & 0.9 \end{bmatrix}$

Hidden MarKov Mocel (HMM)

Example: Forward Motion Prediction

$$\overline{\mathbf{p}}_k = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.5 & 0.1 \\ 0.5 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.5 \\ 0.1 & 0.5 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.2 \\ 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.26 \\ 0.14 \\ 0.14 \\ 0.14 \\ 0.18 \end{bmatrix}.$$



Example: Odd-Face Observation

Predicted Belief (Prior): $\overline{\mathbf{p}}_k = \begin{bmatrix} 0.14 & 0.26 & 0.14 & 0.14 & 0.14 & 0.18 \end{bmatrix}^\top$.

Odd Observation: $z_k = 1 \Rightarrow \mathbf{l}_{z_k=1} = \begin{bmatrix} 0.9 & 0.1 & 0.9 & 0.1 & 0.9 & 0.1 \end{bmatrix}^\top$.

Bayes Update: $\widetilde{\mathbf{p}}_k = \mathbf{l}_{z_k=1} \odot \overline{\mathbf{p}}_k = \begin{bmatrix} 0.126 & 0.026 & 0.126 & 0.014 & 0.126 & 0.018 \end{bmatrix}^\top$.

Normalization constant:

$$\eta_k = \left(\sum_{i=1}^{6} \widetilde{p}_{k,i} \right)^{-1} = \frac{1}{0.126+0.026+0.126+0.014+0.126+0.018} = \frac{1}{0.436}.$$

Posterior Belief: $\mathbf{p}_k = \eta_k \widetilde{\mathbf{p}}_k \approx \begin{bmatrix} 0.2890 & 0.0596 & 0.2890 & 0.0321 & 0.2890 & 0.0413 \end{bmatrix}^\top$.

Bayes Filter

Bayes Filter *is the principal* ***algorithm for calculating the belief*** *in robotics.*

1 **Class** BF
2 **Method** `Prediction`$(bel(\mathbf{x}_{k-1}), \mathbf{u}_k)$
3 **for** *all* $\mathbf{x}_k \in \mathcal{X}$ **do**
4 $\overline{\text{bel}}(\mathbf{x}_k) = \sum_{\mathbf{x}_{k-1} \in \mathcal{X}} p(\mathbf{x}_k \mid \mathbf{u}_k, \mathbf{x}_{k-1}) \text{bel}(\mathbf{x}_{k-1})$;
5 **return** $\overline{\text{bel}}(\mathbf{x}_k)$;
6 **Method** `Update`$(\overline{bel}(\mathbf{x}_k), \mathbf{z}_k)$
7 **for** *all* $\mathbf{x}_k \in \mathcal{X}$ **do**
8 $\text{bel}(\mathbf{x}_k) = \eta p(\mathbf{z}_k \mid \mathbf{x}_k) \overline{\text{bel}}(\mathbf{x}_k)$;
9 **return** $\text{bel}(\mathbf{x}_k)$;
10 **Method** `Filter`$(bel(\mathbf{x}_{k-1}), \mathbf{u}_k, \mathbf{z}_k)$
11 $\overline{\text{bel}}(\mathbf{x}_k) = \text{Prediction}(\text{bel}(\mathbf{x}_{k-1}), \mathbf{u}_k)$;
12 $\text{bel}(\mathbf{x}_k) = \text{Update}(\overline{\text{bel}}(\mathbf{x}_k), \mathbf{z}_k)$;
13 **return** $\text{bel}(\mathbf{x}_k)$;

Sequence Joint Probability

The Recursive Bayes Filter

The power of the probabilistic framework culminates in the **Recursive Bayes Filter**. This algorithm allows a robot to update its belief state $\text{bel}(x_t) = p(x_t \mid z_{1:t}, u_{1:t})$ continuously as it moves and senses. The filter operates in two distinct phases: **Prediction** and **Correction**.

The Prediction Step

Based on the Law of Total Probability and the Markov property, the robot projects its previous belief forward in time using its motion model.

$$\overline{p}(x_t) = \sum_{x_{t-1} \in \mathcal{X}} p(x_t \mid x_{t-1}, u_t) p(x_{t-1})$$

This step typically increases uncertainty (the "blurring" of the PMF), as motion is rarely perfectly deterministic.

The Correction Step

The robot then incorporates the new sensor measurement z_t using Bayes' Rule. We use the normalization constant $\eta = 1/p(z_t)$ to ensure the resulting distribution sums to one.

$$p(x_t \mid z_t) = \eta \, p(z_t \mid x_t) \overline{p}(x_t)$$

This step typically decreases uncertainty (the "sharpening" of the PMF), as the sensor provides evidence about the robot's true location.

Joint PMF of the State Trajectory

The dependency structure allows us to factorize the joint Probability Mass Function (PMF). For a sequence of length k, the joint distribution is:

$$p(\mathbf{x}_{0:k}, \mathbf{u}_{1:k}, \mathbf{z}_{1:k}) = p(\mathbf{x}_0) \prod_{k=1}^{k} p(\mathbf{x}_k \mid \mathbf{x}_{k-1}, \mathbf{u}_k) p(\mathbf{z}_k \mid \mathbf{x}_k)$$

